

High-Level Frameworks

Abstraction That Enables Productivity

Frameworks let you write less code by handling common tasks for you. Instead of building authentication, routing, database connections, and form validation from scratch, you use pre-built components that thousands of developers have already tested and refined.

What Frameworks Provide

Structure — Frameworks impose conventions for organizing code. This means less decision fatigue, easier onboarding for new team members, and codebases that look familiar across projects.

Solved Problems — Security, session management, input validation, CSRF protection—these are hard to get right. Frameworks provide battle-tested implementations.

Ecosystem — Popular frameworks have libraries, plugins, and community resources that extend their capabilities. You rarely start from zero.

Common Framework Categories

Category	Examples	What They Handle
Web Backend	Laravel, Django, Rails, Spring	Routing, ORM, auth, templating
Frontend	React, Vue, Angular	Component rendering, state, DOM updates
Mobile	React Native, Flutter, SwiftUI	Cross-platform UI, device APIs
API	Express, FastAPI, ASP.NET	HTTP handling, middleware, serialization

The Trade-Off

Frameworks impose constraints. You work within their model, use their conventions, and sometimes fight their opinions. This is usually worth it—the productivity gains outweigh the rigidity. But for highly unusual requirements, a framework can become a straitjacket.

Choose frameworks whose opinions match your problem.